Ricardo Castro

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Education

Carnegie Mellon University, Pittsburgh, PA GPA: 3.90
B.S. and M.S. in Mechanical Engineering Expected May 2027
Focus: Mechanical Design, Robotics Hardware & Mechatronics
Coursework: Robot Dynamic Analysis, Intro to CAD/CAE Tools, AI/ML for Manufacturing

Experience

Robotics Research Assistant, Robomechanics Lab, CMU Apr 2025 – Present

- Designed and built Bigfoot, a 1 m quasi-passive biped with one actuated hip by scaling a lab architecture 5× in height and redesigning geometry and mass distribution in Onshape to achieve walking on hardware
- Evaluated foot geometry and actuation in MuJoCo with PD control, then validated on hardware using motion capture and an AK80-8 actuator controlled by a Raspberry Pi 5 over CAN
- Led experiments and manuscript preparation as first author of a submitted conference paper on biped scaling

Mechanical Design Intern, Origami Robotics, Pittsburgh, PA Aug 2025 – Jan 2026

- Designed a teleoperation glove in Onshape with custom 3D-printed finger mechanisms and print-in-place linkages, mapping potentiometer rotation to servo-driven joints for proportional control
- Reduced prototype mass and cost by 50% through parametric redesign in Onshape and FeatureScript

Automation Intern, Nucor, Charleston, SC May 2026 – Aug 2026

- Analyzed roll-stand deflection, loading, and stress in SolidWorks to assess structural performance
- Built a C#/.NET vibration-monitoring application for a hot-mill roll stand, combining signal processing and regression modeling to flag abnormal signatures associated with developing mechanical issues
- Identified crane-orientation near-misses through operator interviews and prototyped a modular IMU/ESP32-S3/LCD HUD attachment to the main remote controller for crane-relative directional feedback

Manufacturing Intern, Manufacturing Futures Institute, CMU May 2024 – Aug 2024

- Designed a modular 30° tilting fixturing table in SolidWorks and machined components using HAAS CNC and water jet with Fusion 360 CAM
- Programmed FANUC-controlled MIG and laser-welding workflows to create repeatable demonstrations

Selected Projects

Bike Carrier System Design CAD/CAE team project

- Led concept selection and FEA-driven redesign of a two-bike hitch rack, reinforcing a critical hitch connection and optimizing material selection to cut assembly mass 66% (80.2 to 27.6 kg)
- Specified a DFM-ready manufacturing plan and per-unit cost buildup for a 45,000-unit production run

Motor-Controlled Scissors Project portfolio

- Built motorized scissors for one-handed cutting by modeling dynamics and tuning PID gains in MATLAB/Simulink, then implementing and validating the Arduino controller across varying loads

Technical Skills

Design & Analysis: SolidWorks, Onshape (FeatureScript), Fusion 360, ANSYS (FEA), GD&T, DFM/DFAM
Fabrication: CNC machining, water jet, 3D printing, laser cutting, manual milling, MIG/laser welding
Programming & Mechatronics: Python, C++, MATLAB/Simulink, MuJoCo, Raspberry Pi, Arduino, CAN
Languages: English (native), Spanish (fluent)

Leadership

Vice President (formerly Treasurer), ASME, CMU Mar 2025 – Present
Managed a $5,000 annual budget and increased event attendance 20% through workshops and seminars

Sets Chair, Greek Sing, Phi Delta Theta Aug 2024 – Present
Led 30+ people to design and build a 20 ft × 12 ft modular stage set with platforms, doors, and moving elements